

Modelling Polynomial Regression to EEG Data

Sreenidhi K C

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1 Introduction

This project investigates the modulation of neural activity in two distinct brain regions—the right auditory cortex and the prefrontal cortex—during a guided meditation task. The objective is to identify the 'true' mathematical model that describes the relationship between the audio stimulus (the meditation guide's voice) and the evoked brain responses. Researchers hypothesise that the auditory cortex exhibits a linear relationship with the audio signal, whereas the prefrontal cortex exhibits a nonlinear relationship. To explore these dependencies, an exploratory data analysis is performed, followed by the evaluation of five candidate nonlinear polynomial regression models to find the optimal mathematical fit.

2 Literature Review

Understanding the distinct functional roles of different brain regions is crucial for modelling neural responses during guided meditation. According to the project guidelines, the right auditory cortex is dedicated to the processing of auditory sensations. In contrast, the prefrontal cortex is responsible for higher-level cognitive processes, specifically relating to executive function, planning, and consciousness.

Based on these functional differences, researchers formulated specific hypotheses regarding how these regions react to auditory stimuli. It is predicted that the auditory cortex is linearly related to the incoming audio signal (the voice of the meditation guide). Conversely, because the prefrontal cortex handles more complex, executive processing, researchers predict a nonlinear relationship with the audio signal.

To mathematically capture and evaluate these complex brain responses, polynomial regression modelling is utilized. The parameter estimation relies on the Least Squares matrix method to minimize the error between the observed signal and the model predictions. However, because all biological signals recorded via electroencephalography (EEG) are subject to distortions, the models must account for additive noise, assumed to be independent and identically distributed (i.i.d) Gaussian with a zero-mean. Therefore, model selection cannot rely on the Residual Sum of Squares (RSS) alone; it must incorporate Information Theory metrics. The Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC) are employed to identify the 'true' model by balancing the goodness-of-fit against the complexity of the model parameters.

3 Methodology

3.1 Data Description

The dataset consists of 2 minutes of electroencephalography (EEG) data sampled at a frequency of 20 Hz. It includes signals measured from the prefrontal cortex (x_1), the auditory cortex (x_2), and the sound signal (y). All signals are assumed to contain additive i.i.d zero-mean Gaussian noise.

3.2 Analysis Pipeline

The methodology followed a structured pipeline to identify the best mathematical model:

1. **Preliminary Data Analysis:** An initial exploratory data analysis investigated the data through time series plots, distribution histograms for each signal, and scatter and correlation plots to examine the dependencies between the audio and brain signals.
2. **Regression Modelling:** Five candidate nonlinear polynomial regression models were evaluated. For each model, parameters (θ) were estimated using the Ordinary Least Squares (OLS) matrix method: $\hat{\theta} = (X^T X)^{-1} X^T y$, utilizing all available data for training.
3. **Model Evaluation:** Models were mathematically compared by computing the Residual Sum of Squared Errors (RSS), the Log-Likelihood function, the AIC, and the BIC. Finally, Quantile-Quantile (Q-Q) plots were analysed to ensure the model prediction errors (residuals) followed a Normal/Gaussian distribution.

3.3 Methodology Flowchart

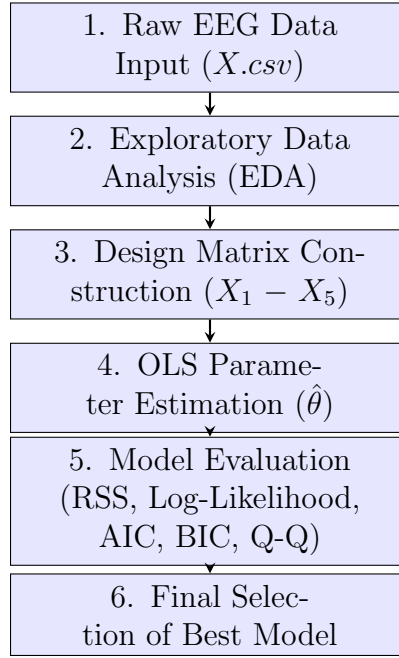


Figure 1: Workflow diagram of the EEG Data Analysis Pipeline.

4 Python Code and Implementation

The core regression modelling was implemented using Python in a Google Colab environment. The primary libraries utilised include **pandas** for dataset loading and **numpy** for constructing the mathematical design matrices and executing the matrix algebra required for parameter estimation. Preliminary data analysis and plotting codes have been omitted for brevity.

```

1 # 1. Import libraries, load dataset, and extract features
2 import numpy as np
3 import pandas as pd

```

```

4 df = pd.read_csv("https://raw.githubusercontent.com/sijuswamy/A-modern-
  approach-to-Research/main/X.csv")
5 x1, x2, y, ones = df["x1"], df["x2"], df["y"], np.ones(len(df["x1"]))
6
7 # 2. Define Design Matrices (X1 to X5)
8 X1 = np.column_stack((x1**3, x2**5, ones))
9 X2 = np.column_stack((x1**4, x2**2, ones))
10 X3 = np.column_stack((x1**3, x2, x1, ones))
11 X4 = np.column_stack((x1, x1**2, x1**3, x2**3, ones))
12 X5 = np.column_stack((x1**3, x1**4, x2, ones))
13
14 # 3. Direct OLS Implementation (Evaluating Model 3)
15 X, n = X3, len(x1)
16 alpha = np.dot((np.dot(np.linalg.inv(np.dot(X.T, X)), X.T)), y)
17
18 # 4. Compute Predictions, RSS, Log-Likelihood, AIC, and BIC
19 yp = np.dot(alpha, X.T)
20 rss = np.sum(np.square(y - yp))
21 llf = -n/2 * np.log(2 * np.pi) - n/2 * np.log(rss/(n - 1)) - (n - 1)/2
22 k = len(alpha)
23 aic, bic = (2 * k) - 2 * llf, k * np.log(n) - 2 * llf

```

Listing 1: Complete Python Implementation for OLS Evaluation

5 Results and Discussions

5.1 Preliminary Data Analysis

The preliminary analysis visualisations strongly supported the researchers' hypotheses. The time series plots verified the continuous integrity of the 2-minute recordings across all features. The scatter plot between the Sound Signal and the Prefrontal Cortex EEG revealed a distinct "S-shaped" curve, visually confirming a strong nonlinear relationship. This dependency was supported by a highly significant correlation coefficient of 0.86. Conversely, the Auditory Cortex plot showed a scattered cloud with a weak correlation of -0.25 , reflecting the hypothesized linear relationship that is heavily obscured by the i.i.d recording noise. Distribution plots successfully validated that the underlying noise is Gaussian, as all three variables exhibited symmetrical, bell-shaped curves.

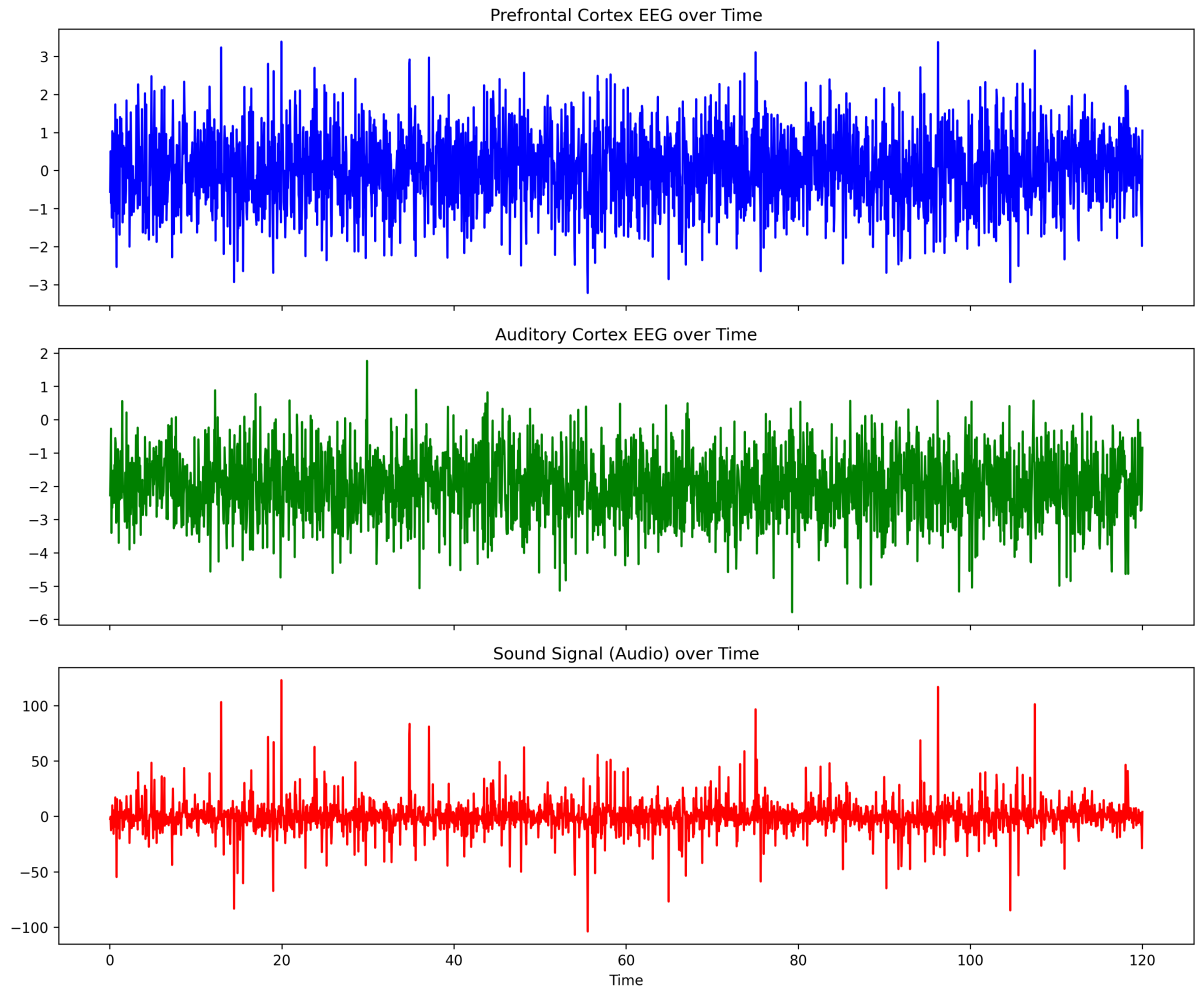


Figure 2: Time series plots of the prefrontal EEG, auditory EEG, and sound signals over the 2-minute recording.

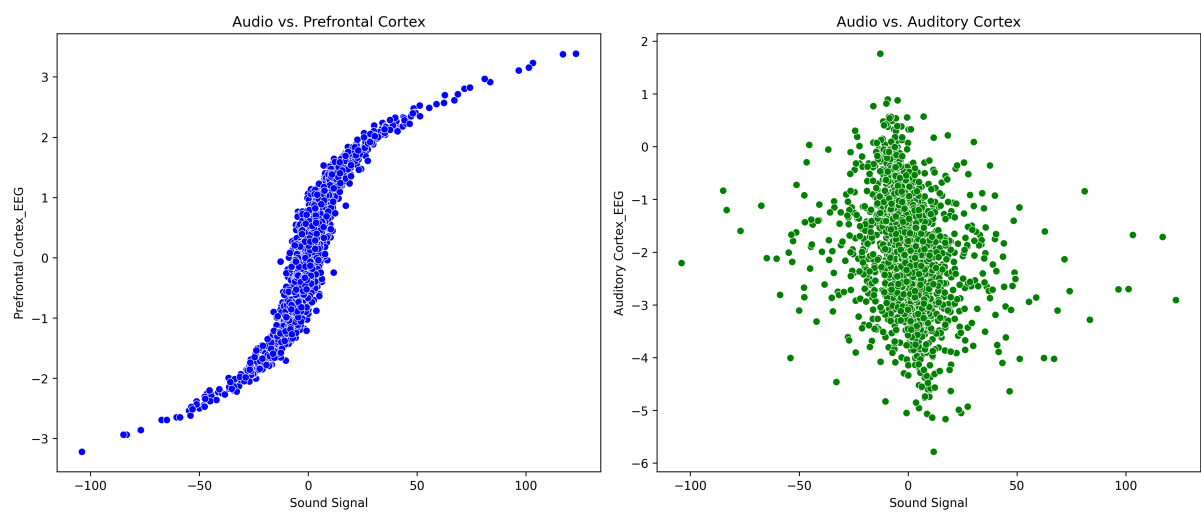


Figure 3: Scatter plots displaying the nonlinear relationship (Prefrontal) and noisy linear relationship (Auditory).

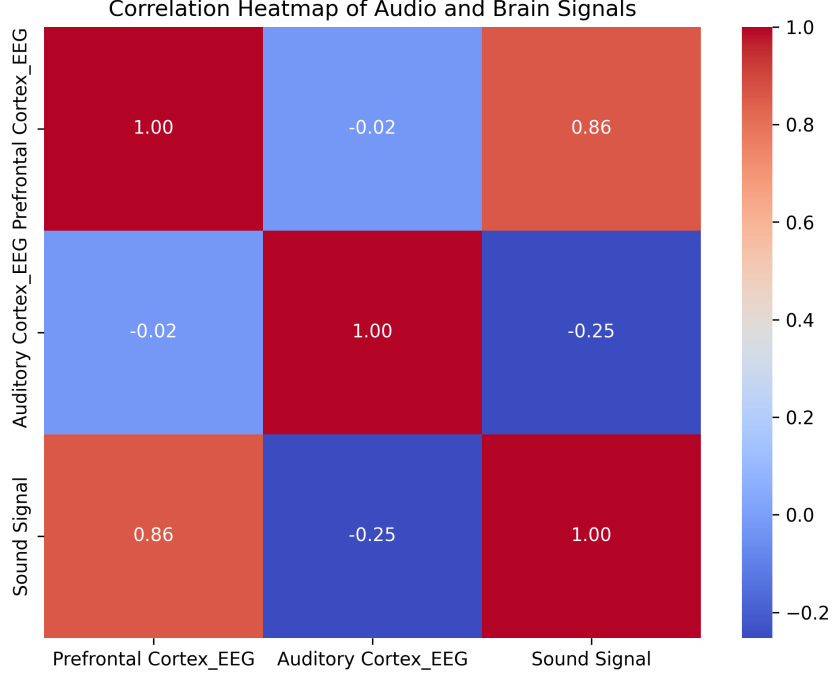


Figure 4: Correlation heatmap displaying the relationship coefficients between the audio and brain signals.

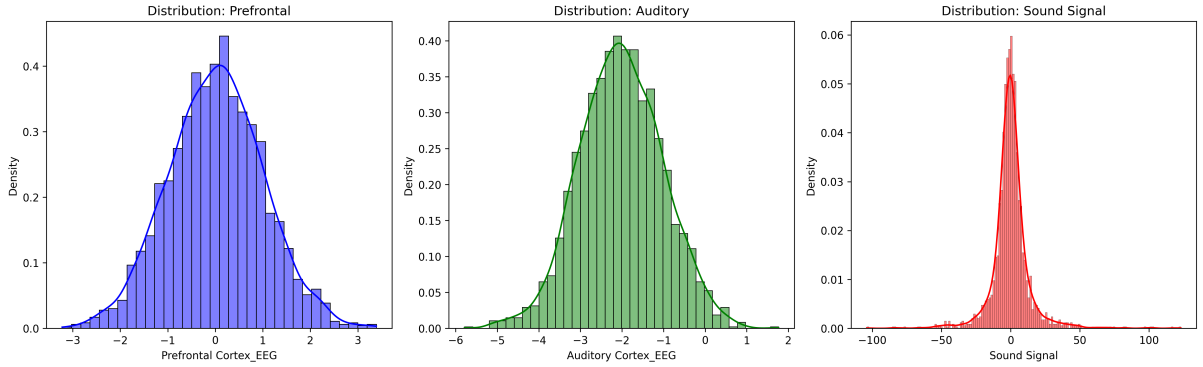


Figure 5: Distribution plots confirming the Gaussian nature of the signals.

5.2 Model Evaluation

To identify the 'true' mathematical relationship between the sound signal (y) and the EEG data from the prefrontal (x_1) and auditory (x_2) cortices, the following five candidate polynomial regression models were evaluated:

$$\textbf{Model 1: } y = \theta_1 x_1^3 + \theta_2 x_2^5 + \theta_{bias} + \epsilon \quad (1)$$

$$\textbf{Model 2: } y = \theta_1 x_1^4 + \theta_2 x_2^2 + \theta_{bias} + \epsilon \quad (2)$$

$$\textbf{Model 3: } y = \theta_1 x_1^3 + \theta_2 x_2 + \theta_3 x_1 + \theta_{bias} + \epsilon \quad (3)$$

$$\textbf{Model 4: } y = \theta_1 x_1 + \theta_2 x_1^2 + \theta_3 x_1^3 + \theta_4 x_2^3 + \theta_{bias} + \epsilon \quad (4)$$

$$\textbf{Model 5: } y = \theta_1 x_1^3 + \theta_2 x_1^4 + \theta_3 x_2 + \theta_{bias} + \epsilon \quad (5)$$

Here, ϵ represents the additive zero-mean Gaussian noise. After estimating the parameters (θ) for all five candidate models using the entire dataset, the mathematical evaluation metrics were compared.

Model	RSS	Log-Likelihood	AIC	BIC
Model 1	30738.22	-6465.49	12936.99	12954.34
Model 2	438230.35	-9654.18	19314.36	19331.71
Model 3	1525.62	-2861.77	5731.54	5754.67
Model 4	7949.27	-4842.58	9695.17	9724.08
Model 5	17138.09	-5764.45	11536.90	11560.04

* Note: RSS = Residual Sum of Squared Errors, AIC = Akaike Information Criterion, BIC = Bayesian Information Criterion.

Table 1: Comparison of evaluation metrics across all 5 candidate models.

Model 3 demonstrated the lowest AIC and BIC scores among all candidates, as well as the most optimal Log-Likelihood value. Because AIC and BIC apply mathematical penalties for overly complex models, these low scores prove that Model 3 provides the best balance: it achieves an excellent fit while effectively avoiding overfitting. Furthermore, the Q-Q plot for Model 3 showed the prediction residuals tracking closely along the straight diagonal line. This indicates that the remaining prediction errors successfully reflect the expected zero-mean Gaussian noise assumption.

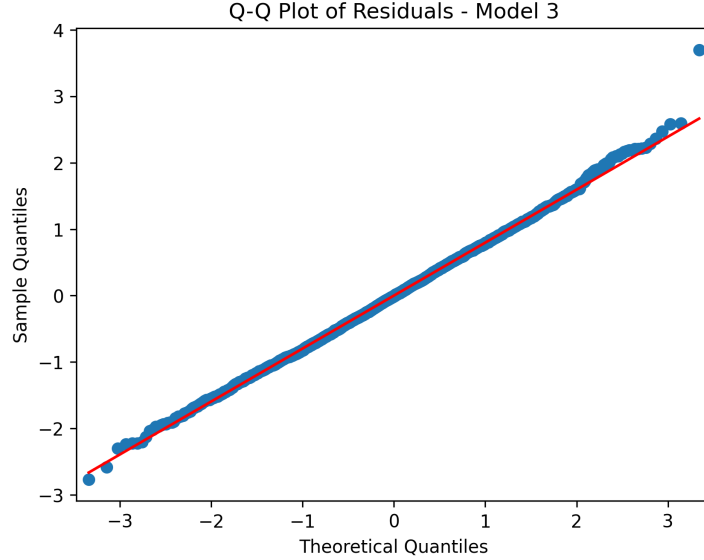


Figure 6: Q-Q plot for Model 3 indicating that residuals follow a Normal/Gaussian distribution.

6 Conclusion

Based on the visual evidence gathered from the exploratory data analysis and the mathematical rigour of the AIC, BIC, and Log-Likelihood evaluation criteria, **Model 3** ($y = \theta_1 x_1^3 + \theta_2 x_2 + \theta_3 x_1 + \theta_{bias}$) is definitively selected as the 'true' regression model for this dataset .

Using the parameters ($\hat{\theta}$) estimated via the Ordinary Least Squares matrix method, the final learned equation is represented as:

$$y = 2.71x_1^3 - 3.15x_2 + 4.18x_1 - 6.65 \quad (6)$$

This mathematical model accurately captures the underlying neurodynamics by applying a linear term to the auditory cortex (x_2) and a nonlinear, cubic formulation to the prefrontal cortex (x_1^3), thereby successfully validating the researchers' initial hypothesis .